Lone Star Math Prep
Number Sense Practice Test • HS 1 • 2026

Final	_____	_____
2nd	_____	_____
1st	_____	_____
	Score	**Initials**

Contestant's Number _____

Read directions carefully before beginning test

DO NOT UNFOLD THIS SHEET UNTIL TOLD TO BEGIN

Directions: Keep this page closed until the person running the test gives the signal to start. You have ten minutes for 80 problems. Work them in order, as quickly and as accurately as you can. EVERY PROBLEM MUST BE SOLVED MENTALLY: no figuring with pencil and paper. Write only the answer, in the blank that follows the problem. A problem marked with a (*) asks for an approximate whole-number answer, and any answer within five percent of the exact value is scored correct. Every other problem requires an exact answer.

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STOP -- WAIT FOR SIGNAL!

(1) $371 - 231 + 9 =$ _____

(2) $4968 - 6437 =$ _____

(3) $3.8(9 + 2) =$ _____

(4) $\frac{7}{9} - \frac{4}{5} =$ _____ (fraction)

(5) $27^2 =$ _____

(6) $72\% =$ _____ (fraction)

(7) $12 \times 75 =$ _____

(8) $12\frac{1}{3}$ minutes = _____ seconds

(9) $612 \div 6 \div 2 =$ _____

***(10)** $76641 + 66326 + 5259 =$ _____

(11) $444 \times \frac{6}{37} =$ _____

(12) MMCMXCIV + DCCII = _____ (Arabic numeral)

(13) The arithmetic mean of 8, 4, 7, 7, 3, 6, 9 and 6 is _____

(14) $\frac{26}{21} + \frac{21}{26} =$ _____ (mixed number)

(15) 0.575757... = _____ (fraction)

(16) If 3 notebooks cost $24.00, then 10 notebooks cost $ _____

(17) If 1 dekagram = 0.4 ounces, then 4.25 ounces = _____ grams

(18) $4405 \times 5 - 25 =$ _____

(19) Each of 80 students does one or both: 40 own a cat and 58 own a dog. The number who only own a cat is _____

***(20)** $367 \times 412 + 934 =$ _____

(21) $0.91666... \times 84 =$ _____

(22) If $5x + 16 = -34$, then $x =$ _____

(23) Forty-five and four-fifths million, fifty-nine thousand, six hundred fifteen = _____ (in digits)

(24) The multiplicative inverse of −2.5 is _____ (decimal)

(25) $6\frac{7}{8} \times 6\frac{1}{8} =$ _____ (mixed number)

(26) If $7^x = 23.8$, then $7^{x-1} =$ _____

(27) $\sqrt{361} + \sqrt{121} =$ _____

(28) $\frac{4}{5}\%$ of 30 is $\frac{3}{8}\%$ of _____

(29) $\frac{3}{8} - \frac{7}{19} =$ _____ (fraction)

***(30)** $336321 \div 13 =$ _____

(31) 13% of $166\frac{2}{3} =$ _____ (mixed number)

(32) $7^2 \div 3.5^2 \times n^2 = 90\frac{1}{4}$, $|n| =$ _____

(33) A square has a perimeter of 108 cm. Its area is _____ sq. cm

(34) The 20th term of 2, 4, 7, 11, 16, ... is 211. The 19th term is _____

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(35) $5\frac{9}{10} \times 10\frac{1}{5} =$ _____ (mixed number)

(36) The product of the roots of $4x^2 - 4x = 5$ is _____

(37) $56\frac{1}{4}\% =$ _____ (fraction)

(38) $2 + 4 + 6 + ... + 76 =$ _____

(39) $66 \times 19.5 =$ _____

*(40) $\sqrt{15089070} =$ _____

(41) $54_6 \times 5_6 - 445_6 =$ _____ 6

(42) If P is $\frac{5}{6}$ of Q and Q is $\frac{5}{8}$ of R, then R is _____ % of P

(43) A regular undecagon with sides of length 14 has a perimeter of _____

(44) The sum of the reciprocals of the first 7 triangular numbers is _____

(45) $(7^{11} + 4^{11}) \div 11$ has a remainder of _____

(46) The sum of the interior angles of a nonagon is _____ °

(47) $76^{17} \div 17$ has a remainder of _____

(48) $(500_6 - 154_6) \times 3_6 =$ _____ 6

(49) $(7^3 - 4^3) \div (7 - 4) =$ _____

*(50) $\sqrt[3]{469205927} =$ _____

(51) $765_9 \times 3_9 =$ _____ 9

(52) $\log_3 4 - \log_3 36 =$ _____

(53) $517_9 \times 6_9 + 557_9 =$ _____ 9

(54) The focus of the parabola $x^2 = 4(y - 1)$ is (0, k). k = _____

(55) $4122_6 \div 11_6$ has a remainder of _____ 6

(56) $\Sigma_{k=1}^{22} (-1)^k k^2 =$ _____

(57) $665_7 \div 6_7$ has a remainder of _____ 7

(58) ${}_6P_4 = k \times {}_6C_4$ and k = _____

(59) $16^4 \times 4^2 \div 64 = 2^k$, k = _____

*(60) $(0.424242... \times 2311)^2 =$ _____

(61) $7 + 15 \times 16 =$ _____

(62) y varies inversely with x and y = 2 when x = 3. When x = 5, y = _____

(63) $\frac{25_7}{60_7} = 0.abbb..._7$ and a + b = _____

(64) $\sin 150° \div \cos(-60°) =$ _____

(65) $444 \times \frac{2}{27} =$ _____ (mixed number)

(66) $\lfloor \sqrt{11} + \sqrt{8} - \sqrt{5} \rfloor =$ _____

(67) $22 + 11 + 5.5 + 2.75 + ... + 0.34375 =$ _____

(68) The 2nd term in the expansion of $(2x + y)^5$ is px^qy^r and p + q + r = _____

(69) $333 \times \frac{27}{37} \times \frac{1}{3} =$ _____

*(70) $(3 + 4 + 5 + ... + 10 + 11 + 12)^2 =$ _____

(71) The directrix of the parabola $y = -2x^2$ is y = _____

(72) If $f(x) = 4x^2 + 2x - 5$, then $f'(-5) =$ _____

(73) If N ÷ 9 has a remainder of 6, then 2N ÷ 9 has a remainder of _____

(74) The vector from (−1, −4) to (−9, 11) has magnitude _____

(75) $0.4202020..._8 =$ _____ 8 (fraction)

(76) $303^3 =$ _____

(77) $\int_2^3 (2x - 4)\,dx \times \int_3^4 (2x - 4)\,dx =$ _____

(78) In the sequence 2, 6, 15, k, 55, 78, 119, ..., k = _____

(79) $\frac{3}{5}$ of 87.5% of 40 = _____

*(80) 8.333...% of 9583 × $\frac{5}{8}$ = _____

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***number) x – y means any integer from x to y is correct**

NOTE: Fractions must be in lowest terms. An answer such as $\frac{2}{3}$ may not be written as a repeating decimal.

(1) 149

(2) −1469

(3) 41.8

(4) $-\frac{1}{45}$

(5) 729

(6) $\frac{18}{25}$

(7) 900

(8) 740

(9) 51

***(10) 140,815 – 155,637**

(11) 72

(12) 3696

(13) 6.25

(14) $2\frac{25}{546}$

(15) $\frac{19}{33}$

(16) 80.00

(17) 106.25

(18) 22000

(19) 22

***(20) 144,532 – 159,744**

(21) 77

(22) −10

(23) 45859615

(24) −0.4

(25) $42\frac{7}{64}$

(26) 3.4

(27) 30

(28) 64

(29) $\frac{1}{152}$

***(30) 24,578 – 27,164**

(31) $21\frac{2}{3}$

(32) 4.75

(33) 729

(34) 191

(35) $60\frac{9}{50}$

(36) −1.25

(37) $\frac{9}{16}$

(38) 1482

(39) 1287

***(40) 3,691 – 4,078**

(41) −3

(42) 192

(43) 154

(44) $\frac{7}{4}$

(45) 0

(46) 1260

(47) 8

(48) 1310

(49) 93

***(50) 739 – 815**

(51) 2516

(52) −2

(53) 4074

(54) 2

(55) 4

(56) 253

(57) 5

(58) 24

(59) 14

***(60) 913,171 – 1,009,293**

(61) 247

(62) 1.2

(63) 4

(64) 1

(65) $32\frac{8}{9}$

(66) 3

(67) 43.65625

(68) 85

(69) 81

***(70) 5,344 – 5,906**

(71) $\frac{1}{8}$

(72) −38

(73) 3

(74) 17

(75) $\frac{103}{176}$

(76) 27818127

(77) 3

(78) 28

(79) 21

***(80) 475 – 524**

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Lone Star Math Prep
Number Sense Practice Test • HS 2 • 2026

	Score	Initials
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2nd		
1st		

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(1) $3654 + 415 =$ _____

(2) $1669 - 675 =$ _____

(3) $218 - 812 =$ _____

(4) $1\frac{5}{6} \times \frac{24}{55} =$ _____ (fraction)

(5) $85 \times 36 =$ _____

(6) $0.0625 =$ _____ (fraction)

(7) $23.6 \times 10^2 - 26 =$ _____

(8) $3! \div 6 \times 17 - 8 + 5 =$ _____

(9) $10 + 14 + 18 + ... + 50 =$ _____

***(10)** $86901 + 13757 + 1352 + 6241 =$ _____

(11) $96 \times 95 =$ _____

(12) CCLII + MMCXI = _____ (Arabic numeral)

(13) $15 \times 360 =$ _____

(14) If tea costs \$0.85 per ounce, then one pound costs \$ _____

(15) $2 \times 3 \times 7 \times 13$ has _____ positive integral divisors

(16) $1 + 2 + 3 + ... + 16 =$ _____

(17) DCCXCII − MCXXVII = _____ (Arabic numeral)

(18) \$2.56 + \$4.50 + \$17.84 = \$ _____

(19) $106 \times 105 =$ _____

***(20)** $30.3 \times 39.1 + 1873 =$ _____

(21) If $2x + 8 = 10$, then $4x - 5 =$ _____

(22) $0.1141414... =$ _____ (fraction)

(23) $2\frac{7}{8} \times 7\frac{2}{3} =$ _____ (mixed number)

(24) $[23 \times 22 + 50] \div 11$ has a remainder of _____

(25) $3\frac{3}{5} \times 2\frac{3}{5} =$ _____ (mixed number)

(26) A = {s,o,l,v,e}, B = {d,o,m,a,i,n} and C = {p,o,l,i,t,e}. The number of distinct elements in $A \cap (B \cap C)$ is _____

(27) $59^2 =$ _____

(28) $46 \times 57 =$ _____

(29) $2793 \times 7 + 49 =$ _____

***(30)** $\sqrt{984780} =$ _____

(31) If $3x + 2y = 3$ and $x - 3y = 10$, then $y =$ _____

(32) The additive inverse of 0.22666... is _____ (fraction)

(33) The number of 4-digit numbers that can be formed from the digits 7, 9, 3, 4, 2 with no digit repeated is _____

(34) The sum of the coefficients in the expansion of $(2x + 3y)^3$ is _____

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(35) $\frac{1}{3} \times 105 =$ _____

(36) If $x = 26$, then $x^2 - 2x + 1 =$ _____

(37) The number of integers k with $4 \le k < 59$ that are relatively prime to 58 is _____

(38) The sum of the roots of $2x^2 + kx - 7 = 0$ is 2. k = _____

(39) $\frac{x+9}{x-5} + \frac{x-5}{x+9} = 2 + \frac{B}{(x+9)(x-5)}$, B = _____

*(40) $\sqrt[3]{939} \times \sqrt{939} \times 939 =$ _____

(41) $3549 \div 91 =$ _____

(42) A regular octahedron has _____ edges

(43) B is a positive integer and $3^B + 2B = 13$. $B^2 =$ _____

(44) $A^9 \times A^5 \div A^{-1} = A^k$, k = _____

(45) $(3^3 + 1^3 - 3) \div 4$ has a remainder of _____

(46) If $\frac{85}{330} = 0.abcbcbc...$, then a + b + c = _____

(47) $(5x + 4y)^2 = ax^2 + bxy + cy^2$ and a + b − c = _____

(48) The vertex of the parabola $y = -3(x - 1)^2 + 2$ is (h, k). h + k = _____

(49) If $4\frac{1}{m} \times n\frac{10}{11} = 12$, where m and n are positive integers, then m + n = _____

*(50) $0.41666... \div 0.3125 \times 479 =$ _____

(51) If $\frac{1}{3} + \frac{1}{6} + \frac{1}{10} + ... + \frac{1}{n} = \frac{3}{4}$, then n = _____

(52) The number of 3-digit numbers less than 600 that can be formed from the digits 9, 1, 4, 6 with no digit repeated is _____

(53) $(6 + 12i) \div i^6 = a + bi$ and a + b = _____

(54) $(7 + i)(5 - 6i) = a + bi$ and a − b = _____

(55) The probability that a positive factor of 75 picked at random is a prime number is _____

(56) If p, q and r are the roots of $2x^3 - 7x^2 - 9x - 6 = 0$, then (p + q + r) + (pq + pr + qr) = _____

(57) $4 + 2 + 1 + 0.5 + ... =$ _____

(58) The point (7, 8) is reflected across the line y = x + 9 to (h, k). h − k = _____

(59) A bag holds 5 black and 5 white socks. Three are drawn without replacement. If the first is white, the odds that the other two are both black are _____

*(60) $125^3 \div 25^4 \times 5^4 =$ _____

(61) If $\begin{pmatrix} 4 & 4 \\ 3 & 6 \end{pmatrix}\begin{pmatrix} 4 & 7 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then b − a = _____

(62) $i^{83} = a\sqrt{b}$ with a and b each 1 or −1, and b − a = _____

(63) The harmonic mean of the roots of $x^3 - 10x^2 + 32x - 32 = 0$ is _____

(64) $402^3 =$ _____

(65) The harmonic mean of the roots of $18x^4 - 171x^3 + 406x^2 - 171x + 18 = 0$ is $\frac{k}{19}$. k = _____

(66) $\lfloor \sqrt{5} + \sqrt{2} \rfloor =$ _____

(67) $(1.5)^{-2} =$ _____ (fraction)

(68) $\begin{vmatrix} 1 & 3 & 3 \\ 1 & 4 & 7 \\ 1 & 5 & 11 \end{vmatrix} =$ _____

(69) $0.13_4 =$ _____ (base-10 fraction)

*(70) A rectangular prism measures 19 in. by 33 in. by 14 in. Its volume is _____ cu. in.

(71) If $f(x) = \frac{3x-5}{7x+5} - 5$, then $f^{-1}(-2) =$ _____

(72) If $f'(x) = -3$ and $f(-1) = 7$, then $f(3) =$ _____

(73) A parallelogram has sides 3 and 14. The sum of the squares of its diagonals is _____

(74) The relative minimum of $f(x) = (x + 10)^{\frac{1}{2}}$ occurs at x = _____

(75) If $f(x) = \frac{3x+13}{2}$, then $f^{-1}(3) =$ _____

(76) $1332_6 \div 2_6 =$ _____ 6

(77) The focus of the parabola $x = y^2 + 2$ is (h, 0). h = _____

(78) $458 \times 452 =$ _____

(79) $10^3 - 7^3 + 4^3 - 1^3 =$ _____

*(80) 6.25% of $6104 \div \frac{7}{12} =$ _____

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*number) x – y means any integer from x to y is correct

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(1) 4069

(2) 994

(3) −594

(4) $\frac{4}{5}$

(5) 3060

(6) $\frac{1}{16}$

(7) 2334

(8) 14

(9) 330

*(10) 102,839 – 113,663

(11) 9120

(12) 2363

(13) 5400

(14) 13.60

(15) 16

(16) 136

(17) −335

(18) $24.90

(19) 11130

*(20) 2,905 – 3,210

(21) −1

(22) $\frac{113}{990}$

(23) $22\frac{1}{24}$

(24) 6

(25) $9\frac{9}{25}$

(26) 1

(27) 3481

(28) 2622

(29) 19600

*(30) 943 – 1,041

(31) $-\frac{27}{11}$

(32) $-\frac{17}{75}$

(33) 120

(34) 125

(35) 35

(36) 625

(37) 26

(38) −4

(39) 196

*(40) 267,677 – 295,853

(41) 39

(42) 12

(43) 4

(44) 15

(45) 1

(46) 14

(47) 49

(48) 3

(49) 10

*(50) 607 – 670

(51) 28

(52) 12

(53) −18

(54) 78

(55) $\frac{1}{3}$

(56) −1

(57) 8

(58) −17

(59) $\frac{5}{13}$

*(60) 2,969 – 3,281

(61) 28

(62) 0

(63) 3

(64) 64964808

(65) 8

(66) 3

(67) $\frac{4}{9}$

(68) 0

(69) $\frac{7}{16}$

*(70) 8,340 – 9,216

(71) $-\frac{10}{9}$

(72) −5

(73) 410

(74) −10

(75) $-\frac{7}{3}$

(76) 444

(77) 2.25

(78) 207016

(79) 720

*(80) 622 – 686

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Number Sense Practice Test • HS 3 • 2026

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(1) $2289 \div 7 =$ _____

(2) $25 \times 66 =$ _____

(3) $\frac{4}{5} \times \frac{5}{6} \times \frac{1}{13} =$ _____ **(fraction)**

(4) $1\frac{1}{2} \times 3\frac{3}{5} =$ _____ **(mixed number)**

(5) $42 \times 55 =$ _____

(6) $32^2 =$ _____

(7) $444 \times \frac{1}{37} =$ _____

(8) $48.89 \times 10^2 + 12 =$ _____

(9) MDCXXVIII − XLVII = _____ **(Arabic numeral)**

***(10)** 46871 + 67821 + 3851 + 7465 = _____

(11) 34 + 55 + 76 + 97 + 118 = _____

(12) The number of integers greater than −32 and less than 23 is _____

(13) CDLXV = _____ **(Arabic numeral)**

(14) $2\frac{14}{45} - \frac{8}{15} =$ _____ **(mixed number)**

(15) $1391 \times 6 + 108 =$ _____

(16) 75% of $\frac{2}{5}$ of 15 = _____

(17) If 1 gram = 0.2 ounces, then 26 ounces = _____ grams

(18) $(6^2 \times 8^2 \times 5) \div (6 \times 5) =$ _____

(19) GCD(72, 16) − LCM(72, 16) = _____

***(20)** $241 \times 219 - 2619 =$ _____

(21) $46 \times 46 =$ _____

(22) $284 \div 25 =$ _____ **(decimal)**

(23) A car goes 198 miles in 4.5 hours. Its average speed is _____ mph

(24) If x = 5, then $x^4 - 10x^2 + 25 =$ _____

(25) $8\frac{4}{5} \times 8\frac{1}{5} =$ _____ **(mixed number)**

(26) 25% of 484 is 50% of _____

(27) $4\frac{1}{5}$ is to 15 as 5 is to _____

(28) 0.1666... = _____ **(fraction)**

(29) The 17th term of 4, 9, 14, 19, ... is 84. The 11th term is _____

***(30)** $\sqrt{542776} =$ _____

(31) $3098 \times 2 + 4 =$ _____

(32) If x + y = −6 and y − x = 2, then xy = _____

(33) If x = 15, then $x^2 - 10x + 25 =$ _____

(34) $\frac{11x-1}{11x+1} + \frac{11x+1}{11x-1} = 2 + \frac{B}{121x^2-1}$, B = _____

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(35) 54 is split into 4 parts in the ratio 1:2:3:4. The largest part is _____

(36) 7 is _____ % of 140

(37) $\frac{41}{45}$ = _____ % (mixed number)

(38) 67_{10} = _____ $_5$

(39) The number of integers greater than 5 and less than 12 that are relatively prime to 12 is _____

*(40) $\sqrt[3]{962818985}$ = _____

(41) 35×75 = _____

(42) $(2^{15} + 8^{15} - 19) \div 10$ has a remainder of _____

(43) $428.00 at 4% simple annual interest for 3 years grows to $ _____

(44) $49^2 + 49$ = _____

(45) The hypotenuse of a 45°-45°-90° triangle is $\sqrt{648}$. Each leg is _____

(46) 2772×16 = _____

(47) $95^{17} \div 17$ has a remainder of _____

(48) $44_7 \times 2_7 + 150_7$ = _____ $_7$

(49) $86^{38} \div 39$ has a remainder of _____

*(50) If 1 Texas league of land is 4428.4 acres, then $12\frac{1}{2}$ leagues = _____ acres

(51) $(5^3 - 8^3) \div (5 - 8)$ = _____

(52) $\log_4 2 + \log_4 128$ = _____

(53) $(7 + 16i) \div i^6 = a + bi$ and $a - b$ = _____

(54) $(7 + 2i)(3 + 5i) = a + bi$ and $a + b$ = _____

(55) $321_4 \div 11_4$ has a remainder of _____ $_4$

(56) $\Sigma_{k=1}^{15} (-1)^k k^2$ = _____

(57) 601^3 = _____

(58) $31210_4 \div 3_4$ has a remainder of _____ $_4$

(59) A bag holds 6 red and 5 green apples. Three are drawn without replacement. If the first is green, the odds that the other two are both red are _____

*(60) A rectangular field is $\frac{11}{16}$ mile by $\frac{7}{8}$ mile. Its area is _____ sq. ft

(61) If $x = 2$ and $y = -1$, then $9x^2 + 2xy + y^2$ = _____

(62) $0.2303030..._5$ = _____ $_5$ (fraction)

(63) $\sqrt{147} + \sqrt{192} = \sqrt{x}$, x = _____

(64) $28^{10} \div 19$ has a remainder of _____

(65) The first four digits after the decimal point of $\frac{23}{60}$ are _____

(66) If $xy = 4$ and $x + y = -7$, then $x^3 + y^3$ = _____

(67) If $xy = -1$ and $x - y = 7$, then $x^3 - y^3$ = _____

(68) $\lceil \sqrt{19} - \sqrt{6} \rceil$ = _____

(69) The polar point $(3, \frac{11\pi}{6})$ has rectangular coordinates (x, y). y = _____

*(70) 2195 miles per hour = _____ feet per second

(71) The remainder when $x^3 + 11x + 11$ is divided by $(x + 3)$ is _____

(72) $\int_{-4}^{4} (2x + 2)\, dx$ = _____

(73) The minimum value of $f(x) = 3x^2 + 8x$ is _____

(74) The slope of the line tangent to $y = 2x^2 + x + 5$ at $x = 3$ is _____

(75) $\int_0^3 \int_0^2 (x + 4)\, dx\, dy$ = _____

(76) $\int_0^2 \int_1^2 xy\, dx\, dy$ = _____

(77) On $1 \le x \le 9$, the maximum value of $f(x) = x + \frac{1}{x}$ is _____

(78) $3.4 - 1.7 + 0.85 - 0.425 + ...$ = _____ (fraction)

(79) $2^3 - 4^3 + 6^3 - 8^3$ = _____

*(80) If 1 Texas vara = $33\frac{1}{3}$ inches, then 28088 yards = _____ varas

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(1) 327

(2) 1650

(3) $\frac{2}{39}$

(4) $5\frac{2}{5}$

(5) 2310

(6) 1024

(7) 12

(8) 4901

(9) 1581

***(10) 119,708 – 132,308**

(11) 380

(12) 54

(13) 465

(14) $1\frac{7}{9}$

(15) 8454

(16) 4.5

(17) 130

(18) 384

(19) −136

***(20) 47,652 – 52,668**

(21) 2116

(22) 11.36

(23) 44

(24) 400

(25) $72\frac{4}{25}$

(26) 242

(27) $\frac{125}{7}$

(28) $\frac{1}{6}$

(29) 54

***(30) 700 – 773**

(31) 6200

(32) 8

(33) 100

(34) 4

(35) 21.6

(36) 5

(37) $91\frac{1}{9}$

(38) 232

(39) 2

***(40) 939 – 1,036**

(41) 2625

(42) 1

(43) 479.36

(44) 2450

(45) 18

(46) 44352

(47) 10

(48) 301

(49) 25

***(50) 52,588 – 58,122**

(51) 129

(52) 4

(53) 9

(54) 52

(55) 2

(56) −120

(57) 217081801

(58) 1

(59) $\frac{1}{2}$

***(60) 15,932,070 – 17,609,130**

(61) 33

(62) $\frac{41}{130}$

(63) 675

(64) 9

(65) 3833

(66) −259

(67) 322

(68) 2

(69) −1.5

***(70) 3,059 – 3,380**

(71) −49

(72) 16

(73) $-\frac{16}{3}$

(74) 13

(75) 30

(76) 3

(77) $\frac{82}{9}$

(78) $\frac{34}{15}$

(79) −352

***(80) 28,819 – 31,851**

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Final _____ _____
2nd _____ _____
1st _____ _____
Score Initials

Contestant's Number _____

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STOP -- WAIT FOR SIGNAL!

(1) $751 + 349 =$ _____

(2) $1106 - 115 =$ _____

(3) $2743 \times 4 =$ _____

(4) $799 \div 9 =$ _____ (mixed number)

(5) $3\frac{3}{4}\% =$ _____ (decimal)

(6) $2.81666... =$ _____ (improper fraction)

(7) $22 + 10 \times 5 - 10 \times 11 =$ _____

(8) $93\frac{3}{4}\% =$ _____ (fraction)

(9) 13.2 is 5% of _____

***(10)** $1874 + 6354 + 7776 =$ _____

(11) $28 + 42 + 56 + 70 + 84 =$ _____

(12) GCD(63, 28) + LCM(63, 28) = _____

(13) 17977987 ÷ 11 has a remainder of _____

(14) If tea costs $1.35 per ounce, then one pound costs $ _____

(15) $53^2 - 46^2 = 9 \times$ _____

(16) $12 \div \frac{3}{4} =$ _____

(17) If 14 pens cost $8.26, then 6 pens cost $ _____

(18) From the end of August 1 to the end of September 25 is _____ days

(19) MMCDXCVII − DCCXII = _____ (Arabic numeral)

***(20)** $492 \times 508 =$ _____

(21) $64 \times 44 =$ _____

(22) $|7 - 6| - |3 - 4| - |3 - 6| - 27 =$ _____

(23) Four million, ninety-one thousand, sixty = _____ (in digits)

(24) The multiplicative inverse of −3.2 is _____ (decimal)

(25) $11\frac{3}{5} \times 11\frac{2}{5} =$ _____ (mixed number)

(26) $109 \times 103 =$ _____

(27) A hexagon has _____ distinct diagonals

(28) $5\frac{2}{5} \div 5\frac{9}{10} =$ _____ (fraction)

(29) $5405 \times 5 - 25 =$ _____

***(30)** $1060928 \div 844 =$ _____

(31) The sum of the coefficients in the expansion of $(5x - 3y)^6$ is _____

(32) $44 \times 46 =$ _____

(33) The reciprocal of $-2\frac{2}{9}$ is _____ (decimal)

(34) The 14th term of 4, 6, 9, 13, 18, ... is 108. The 13th term is _____

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(35) $0.2666... =$ _____ (fraction)

(36) $[38 + 27 \times 4 - 35] \div 6$ has a remainder of _____

(37) If $-3x + y = 28$ and $3x + 5y = 14$, then $y =$ _____

(38) $250_8 =$ _____ 4

(39) $44 \times 43.5 =$ _____

*(40) $\sqrt[3]{487133799} =$ _____

(41) $370_8 =$ _____ 2

(42) If P is $\frac{5}{6}$ of Q and Q is $\frac{1}{4}$ of R, then R is _____ % of P

(43) The coefficient of the x^4y term in the expansion of $(2x + y)^5$ is _____

(44) $82^2 + 22^2 =$ _____

(45) If $5x + y = 10$ and $4x + 3y = 8$, then $y =$ _____

(46) $(6 + 7i)(7 - 9i) = a + bi$ and $a =$ _____

(47) The number of 3-digit numbers greater than 530 that can be formed from the digits 2, 1, 5, 7 with no digit repeated is _____

(48) The product of the roots of $6x^2 - 11x = -10$ is _____

(49) $410_6 \times 4_6 =$ _____ 6

*(50) $6468 \times (112 - 248) =$ _____

(51) The 10th term of 5, 7, 10, 15, 22, 33, ... is 105. The 9th term is _____

(52) $\log_4 8 - \log_4 128 =$ _____

(53) If $f(x) = 5x + \log_5 x$, then $f(25) =$ _____

(54) $1 + 7 + 8 + 15 + 23 + 38 + 61 + 99 + 160 + 259 =$ _____

(55) $5512_7 \div 11_7$ has a remainder of _____ 7

(56) The first four digits after the decimal point of $\frac{41}{60}$ are _____

(57) ${}_5C_3 = k \times {}_5P_2$ and $k =$ _____

(58) $73^2 + 23^2 =$ _____

(59) $3566_9 \div 8_9$ has a remainder of _____ 9

*(60) $125^2 \div 25^6 \times 5^9 =$ _____

(61) $\frac{5 \times 6! + 6 \times 5!}{5!} =$ _____

(62) $\sin^2(\text{Arctan}(-\frac{\sqrt{3}}{3})) =$ _____

(63) Two dice are rolled. The probability that the sum is divisible by 5 is _____ %

(64) $3^3 + k^3 = 756$, $k =$ _____

(65) If $f(x) = x^2 + x - 2$ and $g(x) = x - 6$, then $f(g(4)) =$ _____

(66) $222 \times \frac{4}{27} =$ _____ (mixed number)

(67) $9^{35} \div 22$ has a remainder of _____

(68) If $f(x) = 2x^2 - 3x + 3$ and $g(x) = 3x - 1$, then $f(g(\frac{1}{3})) =$ _____

(69) If $f(x) = 3x + 2$ and $g(x) = 6x - 1$, then $f(g(-\frac{5}{6})) =$ _____

*(70) $\sqrt[4]{7970981907} + 7983 =$ _____

(71) If $f(x) = \frac{2x}{3} + \frac{3}{8}$ and $f^{-1}(x) = ax + b$, then $a =$ _____

(72) If $f(x) = 5x + 6$, then $f^{-1}(f(5) - 6) =$ _____

(73) If $f(x) = 3x^4 + 8x^2 - 3$, then $f''(3) =$ _____

(74) The vector from (4, 2) to (4, 12) has magnitude _____

(75) If $f(x) = \frac{4x + 11}{5}$, then $f^{-1}(14) =$ _____

(76) $8^3 - 6^3 + 4^3 - 2^3 =$ _____

(77) If $\int_0^{2\pi} 4\sin^2 x \, dx = k\pi$, then $k =$ _____

(78) $401^3 =$ _____

(79) $409 \times 207 =$ _____

*(80) A right cylinder has height 24 in. and diameter 6 in. Its volume is _____ cu. cm

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*number) x – y means any integer from x to y is correct

NOTE: Fractions must be in lowest terms. An answer such as $\frac{2}{3}$ may not be written as a repeating decimal.

(1) 1100

(2) 991

(3) 10972

(4) $88\frac{7}{9}$

(5) 0.0375

(6) $\frac{169}{60}$

(7) −38

(8) $\frac{15}{16}$

(9) 264

*(10) 15,204 – 16,804

(11) 280

(12) 259

(13) 5

(14) 21.60

(15) 77

(16) 16

(17) 3.54

(18) 55

(19) 1785

*(20) 237,440 – 262,432

(21) 2816

(22) −30

(23) 4091060

(24) −0.3125

(25) $132\frac{6}{25}$

(26) 11227

(27) 9

(28) $\frac{54}{59}$

(29) 27000

*(30) 1,195 – 1,319

(31) 64

(32) 2024

(33) −0.45

(34) 94

(35) $\frac{4}{15}$

(36) 3

(37) 7

(38) 2220

(39) 1914

*(40) 748 – 826

(41) 11111000

(42) 480

(43) 80

(44) 7208

(45) 0

(46) 105

(47) 8

(48) $\frac{5}{3}$

(49) 2440

*(50) −923,630 – −835,666

(51) 82

(52) −2

(53) 127

(54) 671

(55) 1

(56) 6833

(57) $\frac{1}{2}$

(58) 5858

(59) 4

*(60) 119 – 131

(61) 36

(62) 0.25

(63) $19\frac{4}{9}$

(64) 9

(65) 0

(66) $32\frac{8}{9}$

(67) 1

(68) 3

(69) −16

*(70) 7,868 – 8,695

(71) 1.5

(72) 3.8

(73) 340

(74) 10

(75) 14.75

(76) 352

(77) 4

(78) 64481201

(79) 84663

*(80) 10,564 – 11,675

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	Score	Initials
Final		
2nd		
1st		

Contestant's Number _____

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STOP -- WAIT FOR SIGNAL!

(1) $767 - 2011 =$ _____

(2) $207 \div 9 =$ _____

(3) $1222 + 3758 =$ _____

(4) $4.91 + 4\frac{1}{10} =$ _____ (decimal)

(5) $3.25 =$ _____ (improper fraction)

(6) $30 \times 47 - 15 \times 30 =$ _____

(7) $19 \times 19 =$ _____

(8) $30 + 9 \times 3 - 5 \times 8 =$ _____

(9) $45 \times 45 =$ _____

***(10)** $62731 + 60723 + 2821 =$ _____

(11) $333 \times \frac{6}{37} =$ _____

(12) The sum of the prime divisors of 86 is _____

(13) $57 \times 34 =$ _____

(14) $1638 \times 15 =$ _____

(15) $9 + 10 + 11 + ... + 24 =$ _____

(16) The range of 4, 8, 4, 6, 0, 4, 1, 7, 1 is _____

(17) The multiplicative inverse of 2.625 is _____ (fraction)

(18) $15^2 - 17^2 = 16 \times$ _____

(19) From the beginning of March 1 to the end of May 23 is _____ days

***(20)** $115 \times 221 + 630 =$ _____

(21) The additive inverse of $\frac{2}{21}$ is _____

(22) $x + (x + 1) + (x + 2) + (x + 3) + (x + 4) = 74$. The middle term is _____

(23) A car goes 216 miles in 4.5 hours. Its average speed is _____ mph

(24) $(9^5 + 5^5 + 7) \div 14$ has a remainder of _____

(25) $7\frac{2}{3} \times 7\frac{1}{3} =$ _____ (mixed number)

(26) 4 is to 15 as 30 is to _____

(27) 8.5 pounds of rice at 32 cents per pound cost $ _____

(28) $\frac{1}{x} + \frac{1}{7} = \frac{1}{2}$, x = _____

(29) The area of an equilateral triangle with side 40 is $k\sqrt{3}$. k = _____

***(30)** $\sqrt{400 \times 836} =$ _____

(31) $883_9 =$ _____ 10

(32) $\sqrt[4]{256} =$ _____

(33) A square has an area of 361 sq. cm. Its perimeter is _____ cm

(34) $\frac{7x-2}{7x+2} + \frac{7x+2}{7x-2} = 2 + \frac{B}{49x^2-4}$, B = _____

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(35) The number of positive integers less than 94 that are relatively prime to 94 is _____

(36) $\frac{1}{5} \times 225 =$ _____

(37) If $2x - 3y = 3$ and $3x - 3y = 22$, then $y =$ _____

(38) $22 \times 22 =$ _____

(39) $4\frac{3}{5}\% =$ _____ (fraction)

*(40) 5 yards − 2 feet + 9 inches = _____ centimeters

(41) $303^2 =$ _____

(42) The modulus of $2 + 25i$ is k and $k^2 =$ _____

(43) The point $(-3, -8)$ is reflected across the line $y = 4 - x$ to (h, k). $h - k =$ _____

(44) $1836 \times 14 =$ _____

(45) y varies inversely with x and $y = 6$ when $x = 5$. When $x = 11$, $y =$ _____

(46) $3^3 - 3^2 - 3 =$ _____ 6

(47) $(3x - 9)^2 = ax^2 + bx + c$ and $a + b + c =$ _____

(48) $648_9 \times 6_9 - 327_9 =$ _____ 9

(49) ${}_6P_3 \div {}_6C_1 =$ _____

*(50) $2857.14 \times 38 =$ _____

(51) y varies directly with x and $y = 12$ when $x = 8$. When $y = 6$, $x =$ _____

(52) $\log_3 12 - \log_3 36 =$ _____

(53) $8\frac{2}{m} \times n\frac{7}{37} = 18$, where m and n are natural numbers. $m + n =$ _____

(54) $\frac{8!}{4! \times 2! \times 2!} =$ _____

(55) $\sqrt{a^3} \times \sqrt[6]{a^8} = \sqrt[n]{a^k}$ in lowest terms. $n + k =$ _____

(56) The perimeter of a square increases from 10 cm to 26 cm. Its area increases by _____ sq. cm

(57) $\Sigma_{k=1}^{4} (-1)^k k^2 =$ _____

(58) $\frac{4}{5} + 3 + 3\frac{4}{5} + 6.8 + 10\frac{3}{5} + 17.4 + 28 + 45.4 =$ _____

(59) $65^{15} \div 29$ has a remainder of _____

*(60) $(0.474747... \times 1806)^2 =$ _____

(61) $\sin \frac{11\pi}{6} \times \cos \frac{2\pi}{3} =$ _____

(62) $\lfloor 5\sqrt{3} \rfloor =$ _____

(63) $151 \times 158 =$ _____

(64) $\tan \theta = 1$, θ in quadrant III, $0° \leq \theta < 360°$. $\theta =$ _____ °

(65) If $f(x) = x^2 + 2x + 8$ and $g(x) = 4x - 4$, then $f(g(2)) =$ _____

(66) $888 \times \frac{4}{27} =$ _____ (mixed number)

(67) $10 + 5 + 2.5 + 1.25 + ... =$ _____

(68) If $\begin{vmatrix} -5 & 2 \\ x & -3 \end{vmatrix} = \begin{vmatrix} -4 & -4 \\ 2 & x \end{vmatrix}$, then $x =$ _____

(69) $\sqrt{1034}_5 =$ _____ 5

*(70) A pipe 32 ft long has a diameter of 10 ft. Its volume is _____ cu. ft

(71) The remainder when $x^3 + 2x^2 + 11x - 32$ is divided by $(x + 2)$ is _____

(72) $234 \times 236 =$ _____

(73) If $f(x) = \cos 2x$, then $f''(\frac{\pi}{2}) =$ _____

(74) The maximum point of $y = -x^2 - 8x - 20$ is (a, b). $a + b =$ _____

(75) The circle $x^2 + y^2 = -6y$ has area $k\pi$. $k =$ _____

(76) $\lim_{x \to 0} \frac{\sin 4x}{3x} =$ _____

(77) If $p : q = 5 : 6$ and $p - q = 9$, then $p =$ _____

(78) $30^7 \div 5^2$ has a remainder of _____

(79) In the sequence 1, 1, 5, 4, 12, 9, 22, k, 35, ..., k = _____

*(80) $8.888... \times 37.8 \times 10^3 =$ _____

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***number) x – y means any integer from x to y is correct**

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(1) -1244

(2) 23

(3) 4980

(4) 9.01

(5) $\frac{13}{4}$

(6) 960

(7) 361

(8) 17

(9) 2025

***(10) 119,962 – 132,588**

(11) 54

(12) 45

(13) 1938

(14) 24570

(15) 264

(16) 8

(17) $\frac{8}{21}$

(18) -4

(19) 84

***(20) 24,743 – 27,347**

(21) $-\frac{2}{21}$

(22) 14.8

(23) 48

(24) 7

(25) $56\frac{2}{9}$

(26) 112.5

(27) 2.72

(28) 2.8

(29) 400

***(30) 550 – 607**

(31) 723

(32) 4

(33) 76

(34) 16

(35) 46

(36) 45

(37) $\frac{35}{3}$

(38) 484

(39) $\frac{23}{500}$

***(40) 399 – 440**

(41) 91809

(42) 629

(43) 5

(44) 25704

(45) $\frac{30}{11}$

(46) 23

(47) 36

(48) 3885

(49) 20

***(50) 103,143 – 113,999**

(51) 4

(52) -1

(53) 11

(54) 420

(55) 23

(56) 36

(57) 10

(58) 115.8

(59) 7

***(60) 698,369 – 771,880**

(61) 0.25

(62) 8

(63) 23858

(64) 225

(65) 32

(66) $131\frac{5}{9}$

(67) 20

(68) $-\frac{7}{2}$

(69) 22

***(70) 2,388 – 2,638**

(71) -54

(72) 55224

(73) 4

(74) -8

(75) 9

(76) $\frac{4}{3}$

(77) -45

(78) 0

(79) 16

***(80) 319,200 – 352,800**

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Practice Test 1 — the shortcut for every answer

#	Answer	Shortcut
1	149	Reorder: 371 + 9 − 231.
2	−1469	The other way round, 6437 − 4968 = 1469; the answer is its negative.
3	41.8	Add inside first: 3.8 × 11.
4	$-\frac{1}{45}$	Cross-multiply: $\frac{7\times5-9\times4}{9\times5}=-\frac{1}{45}$.
5	729	$27^2 = 30 \times 24 + 3^2 = 720 + 9$.
6	$\frac{18}{25}$	$\frac{72}{100}$; divide top and bottom by 4.
7	900	×75 is $\times\frac{3}{4}$ then ×100: 12 ÷ 4 = 3, times 3 is 9.
8	740	12 × 60 = 720, plus $\frac{1}{3}$ of 60 = 20.
9	51	One division after the other: 612 ÷ 6 = 102, then ÷ 2 (the same as ÷ 12).
*10	140815 to 155637 (exact: 148226)	Round each number to two figures: 77000 + 66000 + 5300 = 148300.
11	72	111 = 3 × 37, so 444 ÷ 37 = 12, and 12 × 6 = 72.
12	3696	Convert first: 2994 + 702.
13	6.25	The sum is 50; divide by 8.
14	$2\frac{25}{546}$	A fraction plus its reciprocal is 2 + (difference)²/(product): $2+\frac{5^2}{21\times26}$.
15	$\frac{19}{33}$	Two repeating digits go over 99: $\frac{57}{99}$, reduced.
16	80.00	Scale by $\frac{10}{3}$: 24.00 × 10 ÷ 3.
17	106.25	4.25 ÷ 0.4 = 10.625; a dekagram is 10 grams.
18	22000	$25 = 5^2$, so factor out 5: 5 × (4405 − 5) = 5 × 4400.
19	22	They are everyone who does not own a dog: 80 − 58.
*20	144532 to 159744 (exact: 152138)	370 × 410 = 151700, then add about 930.
21	77	0.91666... = $\frac{11}{12}$: 84 ÷ 12 = 7, times 11.
22	−10	5x = −34 − 16 = −50; divide by 5.
23	45859615	Four-fifths of a million is 800 thousand; add the 59 thousand, then three digits for the rest.
24	−0.4	2.5 = $\frac{5}{2}$; flip it and keep the sign.
25	$42\frac{7}{64}$	Same whole part, fractions sum to 1: 6 × 7 = 42, then $\frac{7}{8}\times\frac{1}{8}$.
26	3.4	The exponent drops by 1, so divide by 7: 23.8 ÷ 7.
27	30	$\sqrt{361}$ = 19 and $\sqrt{121}$ = 11.
28	64	The percent signs cancel: $\frac{4}{5}\times30\div\frac{3}{8}=\frac{4}{5}\times30\times\frac{8}{3}$.
29	$\frac{1}{152}$	Cross-multiply: $\frac{3\times19-8\times7}{8\times19}=\frac{1}{152}$.
*30	24578 to 27164 (exact: ≈ 25870.8)	Keep three figures of each: 336000 ÷ 13 ≈ 25846.
31	$21\frac{2}{3}$	$166\frac{2}{3}=\frac{500}{3}$, so the 100s cancel: $13\times\frac{5}{3}=\frac{65}{3}$.
32	4.75	7 ÷ 3.5 = 2, so $(2n)^2 = 90\frac{1}{4} = 9.5^2$ and |n| = 9.5 ÷ 2.
33	729	The side is 108 ÷ 4 = 27; square it.
34	191	The differences are 2, 3, 4, ..., so the step into the 20th term is 20: 211 − 20.
35	$60\frac{9}{50}$	All four products are easy: $5\times10+5\times\frac{1}{5}+10\times\frac{9}{10}+\frac{9}{10}\times\frac{1}{5}=50+1+9+\frac{9}{50}$.

#	Answer	Shortcut
36	−1.25	With everything on one side the constant is −5; the product is c ÷ a = −5 ÷ 4.
37	$\frac{9}{16}$	$\frac{1}{16} = 6\frac{1}{4}\%$, and this is 9 of them.
38	1482	The first n even numbers add up to n(n + 1): 38 × 39.
39	1287	Halve 66 and double 19.5: 33 × 39, a ×11 product.
*40	3691 to 4078 (exact: ≈ 3884.5)	$38^2 = 1444$ and $39^2 = 1521$ bracket the leading digits 1508, so the root is about 3900.
41	−3	Multiply first; the product is the smaller number, so the answer is negative. In base 10: 34 × 5 − 173 = −3, which is -3_6.
42	192	$P = \frac{5}{6} \times \frac{5}{8} \times R = \frac{25}{48} \times R$, so $R = \frac{48}{25} \times P$.
43	154	An undecagon has 11 sides: 11 × 14.
44	$\frac{7}{4}$	Each reciprocal is 2/n − 2/(n + 1), so the sum telescopes to $2 - \frac{2}{8}$.
45	0	With an odd exponent $7^{11} + 4^{11}$ is divisible by 7 + 4 = 11, so nothing is left over.
46	1260	(n − 2) × 180 = 7 × 180.
47	8	Fermat's little theorem: $a^p \equiv a \pmod p$ for a prime p, so reduce 76 mod 17.
48	1310	Do the parentheses first, carrying 6s. In base 10: (180 − 70) × 3 = 330, which is 1310_6.
49	93	$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, so the quotient is 49 + 28 + 16.
*50	739 to 815 (exact: ≈ 777.1)	$77^3 = 456533$ and $78^3 = 474552$ bracket the leading digits 469205, so the root is about 780.
51	2516	Multiply right to left, carrying 9s. In base 10: 626 × 3 = 1878, which is 2516_9.
52	−2	It is the log of the quotient: 4 ÷ 36 = $\frac{1}{9} = 3^{-2}$.
53	4074	Multiply first, then add, all in base 9. In base 10: 421 × 6 + 457 = 2983, which is 4074_9.
54	2	4p = 4, so p = 1; the focus is p from the vertex: 1 + 1.
55	4	11_6 is one more than the base, so alternate signs on the digits from the right: 2 − 2 + 1 − 4 = −3, which leaves 4 on division by 7.
56	253	Pairing the terms leaves 1 + 2 + ... + 22 = 253 with the sign of the last term.
57	5	Dividing by 6 in base 7 works like casting out nines: the digits add to 17, and 17 ÷ 6 leaves 5.
58	24	${}_6P_4 = 4! \times {}_6C_4$.
59	14	Everything as a power of 2: 4 × 4 + 2 × 2 − 6.
*60	913171 to 1009293 (exact: ≈ 961231.7)	0.424242... = $\frac{14}{33}$: $\frac{14}{33}$ of 2311 is about 980, and its square is about 961232.
61	247	Multiply first: 15 × 16 = 30 × 8 = 240, then add 7.
62	1.2	xy stays 6: 6 ÷ 5.
63	4	In base 10 the fraction is $\frac{19}{42}$; multiplying by 7 again and again peels off the digits 3, 1, 1, ...
64	1	$\sin 150° = \frac{1}{2}$ and $\cos(-60°) = \frac{1}{2}$.
65	$32\frac{8}{9}$	111 = 3 × 37, so the 3 cancels into 27: $148 \times \frac{2}{9} = \frac{296}{9}$.
66	3	$\sqrt{11} \approx 3.32$, $\sqrt{8} \approx 2.83$, $\sqrt{5} \approx 2.24$: the total is about 3.91.
67	43.65625	With ratio $\frac{1}{2}$ the sum is twice the first term minus the last term: 44 − 0.34375.
68	85	The term is $80x^4y$, and the exponents always add to 5.
69	81	333 = 37 × 9: the 37s cancel and 9 ÷ 3 = 3, leaving 3 × 27 × 1.
*70	5344 to 5906 (exact: 5625)	10 terms averaging 7.5 add up to 75, and 75^2 is about 5600.
71	$\frac{1}{8}$	In $x^2 = 4py$, $4p = -\frac{1}{2}$, so $p = -\frac{1}{8}$; the directrix is y = −p.

#	Answer	Shortcut
72	−38	f′(x) = 8x + 2; substitute x = −5.
73	3	Multiply the remainder: 2 × 6 = 12, which leaves 3.
74	17	The components are −8 and 15: the 8, 15, 17 right triangle.
75	$\frac{103}{176}$	As with repeating decimals, working in base 8: $\frac{420 - 4}{770} = \frac{414}{770}$, which reduces to $\frac{103}{176}$.
76	27818127	$(300 + 3)^3 = 300^3 + 3 \times 300^2 \times 3 + 3 \times 300 \times 3^2 + 3^3 = 27000000 + 810000 + 8100 + 27$.
77	3	The antiderivative is $x^2 - 4x$: the integrals are 1 and 3; a product cannot be merged.
78	28	Term n is n times the n-th prime: 4 × 7.
79	21	87.5% = $\frac{7}{8}$: $\frac{3}{5} \times \frac{7}{8} \times 40$, cancelling first.
*80	475 to 524 (exact: ≈ 499.1)	8.333...% = $\frac{1}{12}$: 9583 ÷ 12 ≈ 799, then × $\frac{5}{8}$ gives about 499.

Practice Test 2 — the shortcut for every answer

#	Answer	Shortcut
1	4069	Add in pieces: 3654 + 400 = 4054, then + 15.
2	994	1669 − 669 is exactly 1000; taking 6 more leaves 994.
3	−594	A 3-digit number minus its reversal is 99 × (difference of the end digits): 99 × (2 − 8).
4	$\frac{4}{5}$	$1\frac{5}{6} = \frac{11}{6}$; 6 cancels into 24 and 11 into 55, leaving $\frac{4}{5}$.
5	3060	Halve 36 and double 85: 18 × 170.
6	$\frac{1}{16}$	0.0625 = $\frac{1}{16}$, and 0.0625 is 1 of them.
7	2334	Move the decimal point 2 places to get 2360, then subtract 26.
8	14	3! = 6; × and ÷ before + and −: 17 − 8 + 5 = 14.
9	330	(50 − 10) ÷ 4 + 1 = 11 terms, so 11 × (10 + 50) ÷ 2.
*10	102839 to 113663 (exact: 108251)	Round each number to two figures: 87000 + 14000 + 1400 + 6200 = 108600.
11	9120	Near 100: 96 − 5 = 91, then (−4) × (−5) = 20: 9100 + 20.
12	2363	Convert first: 252 + 2111.
13	5400	×15: 360 plus half of it is 540, then ×10.
14	13.60	A pound is 16 ounces: double 0.85 four times.
15	16	In prime factors it is 2 × 3 × 7 × 13; add 1 to each exponent and multiply.
16	136	n(n + 1)/2 = 16 × 17 ÷ 2.
17	−335	792 − 1127: subtract the other way and make it negative.
18	$24.90	Work in cents: 256 + 450 + 1784 = 2490.
19	11130	Near 100: 106 + 5 = 111, then 6 × 5 = 30: 11100 + 30.
*20	2905 to 3210 (exact: ≈ 3057.7)	About 30 × 39 = 1170, plus 1900.
21	−1	x = 1, so 4x − 5 = 4 − 5.
22	$\frac{113}{990}$	(114 − 1)/990 = $\frac{113}{990}$, reduced.
23	$22\frac{1}{24}$	Both are 23 over something: $\frac{23}{8} \times \frac{23}{3} = \frac{23^2}{24} = \frac{529}{24}$.
24	6	Use remainders on division by 11: 1 × 0 + 6 = 6.
25	$9\frac{9}{25}$	3 × 2 = 6, then $\frac{3}{5} \times (3 + 2) = 3$, then $\frac{3}{5}$ squared.
26	1	A ∩ (B ∩ C) = {o}.
27	3481	Near 50: 25 + 9 = 34, then $9^2 = 81$.
28	2622	Distances from 50 are −4 and 7: 2500 + 50 × 3 + (−4) × 7 = 2500 + 150 − 28.
29	19600	49 = 7^2, so factor out 7: 7 × (2793 + 7) = 7 × 2800.
*30	943 to 1041 (exact: ≈ 992.4)	$99^2 = 9801$ and $100^2 = 10000$ bracket the leading digits 9847, so the root is about 990.
31	$-\frac{27}{11}$	Eliminate x: −11y = 27.
32	$-\frac{17}{75}$	0.22666... = $\frac{17}{75}$; then change the sign.
33	120	Fill the places left to right: 5 × 4 × 3 × 2.
34	125	Set x = y = 1: $(2 + 3)^3 = 5^3$.
35	35	105 ÷ 3 = 35.
36	625	The trinomial is $(x - 1)^2$, and x − 1 = 25.
37	26	28 of the integers from 1 to 58 are relatively prime to 58 (58 = 2 × 29); 2 of those are no larger than 3.
38	−4	The sum of the roots is −k ÷ 2, so k = −2 × (2).

#	Answer	Shortcut
39	196	$p/q + q/p = 2 + (p - q)^2/(pq)$, and here $p - q = 9 + 5 = 14$.
*40	267677 to 295853 (exact: ≈ 281764.9)	$\sqrt[3]{939} \approx 9.8$ and $\sqrt{939} \approx 30.6$; $9.8 \times 30.6 \times 939$ is about 281600.
41	39	Divide left to right, then check: $91 \times 39 = 3549$.
42	12	8 faces × 3 sides ÷ 2, since two faces share each edge.
43	4	Try small values: $B = 2$ gives 13, so $B^2 = 4$.
44	15	Add exponents to multiply, subtract to divide: $9 + 5 - (-1) = k$.
45	1	With an odd exponent $3^3 + 1^3$ is divisible by $3 + 1 = 4$, so only -3 matters: it leaves 1.
46	14	$\frac{85}{330} = \frac{255}{990}$, and 255 ÷ 99 = 2 remainder 57: 0.25757...
47	49	$a = 25$, $b = 40$ and $c = 16$.
48	3	Vertex form $y = a(x - h)^2 + k$: $h = 1$ (the sign flips) and $k = 2$.
49	10	Both mixed numbers must cancel: $4\frac{1}{8} = \frac{33}{8}$ and $2\frac{10}{11} = \frac{32}{11}$, and $\frac{33}{8} \times \frac{32}{11} = 12$; so $m = 8$ and $n = 2$.
*50	607 to 670 (exact: ≈ 638.7)	$0.41666... = \frac{5}{12}$ and $0.3125 = \frac{5}{16}$, so the quotient is $\frac{16}{12} = \frac{4}{3}$, and $\frac{4}{3} \times 479 \approx 639$.
51	28	The terms are reciprocals of triangular numbers and through $1/T_k$ they add up to $1 - 2/(k + 1) = \frac{3}{4}$, so $k = 7$ and $n = T_7$.
52	12	Count by leading digit (1: 6, 4: 6, 6: 0, 9: 0).
53	−18	$i^6 = -1$, so dividing by it multiplies by −1: the quotient is $-6 - 12i$.
54	78	$a = 7 \times 5 - 1 \times (-6) = 41$ and $b = 7 \times (-6) + 1 \times 5 = -37$.
55	$\frac{1}{3}$	3, 5 qualify: 2 of the 6.
56	−1	From the coefficients: sum 3.5, sum of pairs −4.5, product 3.
57	8	First term ÷ (1 − ratio): $4 \div \frac{1}{2} = 4 \times 2$.
58	−17	Solve the line for each variable: $h = y - 9$ and $k = x + 9$. The image is (−1, 16).
59	$\frac{5}{13}$	With one white gone, $P = \frac{5}{9} \times \frac{4}{8} = \frac{5}{18}$, so the odds are 5 to 13.
*60	2969 to 3281 (exact: 3125)	Everything as a power of 5: $9 - 8 + 4 = 5$, and $5^5 = 3125$.
61	28	Row times column for only what is asked: $b = 48$, $a = 20$.
62	0	$83 \div 4$ leaves 3, so $i^{83} = -i$, which is $-1\sqrt{-1}$.
63	3	3 ÷ (sum of reciprocals) = 3 × product ÷ (sum of pairs) = 3 × 32 ÷ 32.
64	64964808	$(400 + 2)^3 = 400^3 + 3 \times 400^2 \times 2 + 3 \times 400 \times 2^2 + 2^3 = 64000000 + 960000 + 4800 + 8$.
65	8	4 ÷ (sum of reciprocals) = 4 × (constant term) ÷ (opposite of the x coefficient) = 4 × 18 ÷ 171.
66	3	$\sqrt{5} \approx 2.24$, $\sqrt{2} \approx 1.41$: the total is about 3.65.
67	$\frac{4}{9}$	$1.5 = \frac{3}{2}$, and the negative exponent flips it: 2^2 over 3^2.
68	0	Subtract row 1 from the others, leaving the 2×2 determinant $1 \times 8 - 4 \times 2$.
69	$\frac{7}{16}$	0.13 in base 4 is $\frac{1}{4} + \frac{3}{16} = \frac{7}{16}$.
*70	8340 to 9216 (exact: 8778)	$19 \times 33 = 627$, then $627 \times 14 \approx 630 \times 14$.
71	$-\frac{10}{9}$	The fraction must equal 3: cross-multiply to $3x - 5 = 3(7x + 5)$ and solve for x.
72	−5	f is a line of slope −3: $7 + (-3) \times 4$.
73	410	Parallelogram law: $2(3^2 + 14^2) = 2 \times 205$.
74	−10	A square root is smallest where it starts, at $x + 10 = 0$.
75	$-\frac{7}{3}$	Set $f(x) = 3$: $3x + 13 = 6$, so $x = -\frac{7}{3}$.

#	Answer	Shortcut
76	444	Short division, bringing down 6s instead of tens. In base 10: 344 ÷ 2 = 172, which is 444_6.
77	2.25	The vertex is at x = 2 and 4p = 1, so the focus is $\frac{1}{4}$ past it: 2 + $\frac{1}{4}$.
78	207016	Same lead 45, ones add to 10: 45 × 46 = 2070, then 8 × 2 = 16.
79	720	Cubes: 1000 − 343 + 64 − 1.
*80	622 to 686 (exact: 654)	6.25% = $\frac{1}{16}$: 6104 ÷ 16 ≈ 382, then ÷ $\frac{7}{12}$ gives about 654.

Practice Test 3 — the shortcut for every answer

#	Answer	Shortcut
1	327	Short division, left to right; check with $7 \times 327 = 2289$.
2	1650	×25 is ÷4 then ×100: $66 \div 4 = 16.5$.
3	$\frac{2}{39}$	Cancel across before multiplying (a factor of 10 goes), leaving $\frac{2}{39}$.
4	$5\frac{2}{5}$	Improper fractions: $\frac{3}{2} \times \frac{18}{5} = \frac{27}{5}$.
5	2310	Halve 42 and double 55: 21×110.
6	1024	$32^2 = 30 \times 34 + 2^2 = 1020 + 4$.
7	12	$111 = 3 \times 37$, so $444 \div 37 = 12$; then $\times 1$.
8	4901	Move the decimal point 2 places to get 4889, then add 12.
9	1581	Convert first: $1628 - 47$.
*10	119708 to 132308 (exact: 126008)	Round each number to two figures: $47000 + 68000 + 3900 + 7500 = 126400$.
11	380	Evenly spaced, so $5 \times$ the middle term: 5×76.
12	54	$23 - (-32) - 1$: 31 negatives, zero and 22 positives.
13	465	A smaller numeral in front of a larger one subtracts: CM = 900, XL = 40, IV = 4.
14	$1\frac{7}{9}$	$\frac{8}{15} = \frac{24}{45}$; borrow $1 = \frac{45}{45}$, so the fraction is $\frac{45 + 14 - 24}{45} = \frac{35}{45}$, then reduce.
15	8454	$1391 \times 6 = 8400 - 54 = 8346$; then add 108.
16	4.5	$\frac{2}{5}$ of 15 is 6; then $75 \times 6 = 450$, two decimal places.
17	130	$26 \div 0.2 = 130$.
18	384	Cancel 6×5 out of the product before multiplying it out; what is left is 384.
19	−136	GCD = 8 and LCM = $72 \times 16 \div 8 = 144$.
*20	47652 to 52668 (exact: 50160)	$240 \times 220 = 52800$, then subtract about 2600.
21	2116	Near 50: $25 - 4 = 21$, then $4^2 = 16$.
22	11.36	÷25 is ×4 then ÷100: $284 \times 4 = 1136$.
23	44	Double both: $396 \div 9$.
24	400	It is $(x^2 - 5)^2 = 20^2$.
25	$72\frac{4}{25}$	Same whole part, fractions sum to 1: $8 \times 9 = 72$, then $\frac{4}{5} \times \frac{1}{5}$.
26	242	The percents cancel: $484 \times \frac{25}{50} = 484 \times \frac{1}{2}$.
27	$\frac{125}{7}$	$15 \times 5 \div \frac{21}{5} = 75 \times \frac{5}{21}$.
28	$\frac{1}{6}$	$(16 - 1)/90 = \frac{15}{90}$, reduced.
29	54	The common difference is 5; step back 6×5 from 84.
*30	700 to 773 (exact: ≈ 736.7)	$73^2 = 5329$ and $74^2 = 5476$ bracket the leading digits 5427, so the root is about 740.
31	6200	$4 = 2^2$, so factor out 2: $2 \times (3098 + 2) = 2 \times 3100$.
32	8	$((x + y)^2 - (x - y)^2) \div 4 = (36 - 4) \div 4$.
33	100	The trinomial is $(x - 5)^2$, and $x - 5 = 10$.
34	4	The numerator is $2(121x^2 + 1) = 2(121x^2 - 1) + 4 \times 1$.
35	21.6	The ratio terms add to 10, so the part is $\frac{4}{10}$ of 54.
36	5	$7 \div 140 = 0.05$.
37	$91\frac{1}{9}$	$\frac{41}{45} \times 100 = \frac{820}{9}$; write it as a mixed number.
38	232	Take out powers of 5: $67 = 2{\times}25 + 3{\times}5 + 2$.

#	Answer	Shortcut
39	2	4 of the integers from 1 to 12 are relatively prime to 12 ($12 = 2^2 \times 3$); 2 of those are no larger than 5.
*40	939 to 1036 (exact: ≈ 987.4)	$98^3 = 941192$ and $99^3 = 970299$ bracket the leading digits 962818, so the root is about 990.
41	2625	Tens digits add to 10, same ones digit: $3 \times 7 + 5 = 26$, then $5^2 = 25$.
42	1	With an odd exponent $2^{15} + 8^{15}$ is divisible by $2 + 8 = 10$, so only -19 matters: it leaves 1.
43	479.36	$I = Prt = 428 \times 0.04 \times 3 = 51.36$; add the principal.
44	2450	Factor: $49 \times (49 + 1) = 49 \times 50$.
45	18	$\sqrt{648} = 18\sqrt{2}$, so each leg is 18.
46	44352	$2772 \times 10 + 2772 \times 6 = 27720 + 16632$.
47	10	Fermat's little theorem: $a^p \equiv a \pmod p$ for a prime p, so reduce 95 mod 17.
48	301	Multiply first, then add, all in base 7. In base 10: $32 \times 2 + 84 = 148$, which is 301_7.
49	25	$86 \equiv 8 \pmod{39}$; $8^4 \equiv 1 \pmod{39}$, so only the exponent's remainder 38 mod 4 = 2 matters: 8^2 leaves 25.
*50	52588 to 58122 (exact: 55355)	$12\frac{1}{2} \times 4400 = 55000$.
51	129	$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, so the quotient is $25 + 40 + 64$.
52	4	It is the log of the product: $2 \times 128 = 256 = 4^4$.
53	9	$i^6 = -1$, so dividing by it multiplies by -1: the quotient is $-7 - 16i$.
54	52	$a = 7{\times}3 - 2{\times}5 = 11$ and $b = 7{\times}5 + 2{\times}3 = 41$.
55	2	11_4 is one more than the base, so alternate signs on the digits from the right: $1 - 2 + 3 = 2$, which leaves 2 on division by 5.
56	−120	Pairing the terms leaves $1 + 2 + \ldots + 15 = 120$ with the sign of the last term.
57	217081801	$(600 + 1)^3 = 600^3 + 3 \times 600^2 \times 1 + 3 \times 600 \times 1^2 + 1^3 = 216000000 + 1080000 + 1800 + 1$.
58	1	Dividing by 3 in base 4 works like casting out nines: the digits add to 7, and $7 \div 3$ leaves 1.
59	$\frac{1}{2}$	With one green gone, $P = \frac{6}{10} \times \frac{5}{9} = \frac{1}{3}$, so the odds are 1 to 2.
*60	15932070 to 17609130 (exact: 16770600)	$\frac{7}{8}$ mile is 7×660 ft and $\frac{11}{16}$ mile is 11×330 ft, so the area is $77 \times 217800 \approx 77 \times 218$ thousand.
61	33	$9 \times 4 + 2 \times (-2) + 1$.
62	$\frac{41}{130}$	As with repeating decimals, working in base 5: $\frac{230 - 2}{440} = \frac{223}{440}$, which reduces to $\frac{41}{130}$.
63	675	$\sqrt{147} = 7\sqrt{3}$ and $\sqrt{192} = 8\sqrt{3}$; together $15\sqrt{3} = \sqrt{225 \times 3}$.
64	9	$28 \equiv 9 \pmod{19}$, and $9^9 \equiv 1$ ($a^{(p-1)/2}$ is always ±1 for a prime p); one more factor of 9 gives 9.
65	3833	$\frac{23}{60} = 0.38333\ldots$: the digits settle into a repeating one.
66	−259	$(x + y)^3 - 3xy(x + y) = (-343) - 3(4)(-7)$.
67	322	$(x - y)^3 + 3xy(x - y) = 343 + 3(-1)(7)$.
68	2	$\sqrt{19} \approx 4.36$, $\sqrt{6} \approx 2.45$: the difference is about 1.91, and the ceiling rounds up.
69	−1.5	$y = r \sin\theta = 3 \times (-\frac{1}{2})$.
*70	3059 to 3380 (exact: ≈ 3219.3)	60 mph = 88 ft/s, so multiply by $\frac{22}{15}$, a little under 1.5: about 3219.
71	−49	Remainder theorem: substitute $x = -3$, working synthetically.
72	16	The odd power cancels over a symmetric interval, leaving 2×8.
73	$-\frac{16}{3}$	$c - b^2/(4a) = 0 - \frac{64}{12}$.
74	13	$y' = 4x + 1$ at $x = 3$.

#	Answer	Shortcut
75	30	No y in the integrand, so it is 3 times the integral of x + 4 from 0 to 2.
76	3	It splits into a product: $(2^2 - 1^2)/2 \times (2^2 - 0^2)/2$.
77	$\frac{82}{9}$	f increases for $x \geq 1$, so use the endpoints: $f(9) = \frac{82}{9}$.
78	$\frac{34}{15}$	The ratio is $-\frac{1}{2}$: first term $\div (1 + \frac{1}{2}) = 3.4 \times \frac{2}{3}$.
79	−352	Cubes: 8 − 64 + 216 − 512.
*80	28819 to 31851 (exact: ≈ 30335.0)	$36 \div 33\frac{1}{3} = \frac{27}{25} = 1.08$: add 8% to 28088.

Practice Test 4 — the shortcut for every answer

#	Answer	Shortcut
1	1100	The last two digits, 51 and 49, make 100: 7 + 3 + 1 = 11 hundreds.
2	991	1106 − 106 is exactly 1000; taking 9 more leaves 991.
3	10972	Digit by digit from the left: 2743 × 4 = 10972.
4	$88\frac{7}{9}$	799 = 9 × 88 + 7, so put the remainder 7 over 9.
5	0.0375	$3\frac{3}{4} = 3.75$; move the decimal point two places left.
6	$\frac{169}{60}$	$(816 - 81)/900 = \frac{49}{60}$; with the 2, that is $\frac{169}{60}$.
7	−38	× and ÷ left to right before + and −: 22 + 50 − 110 = −38.
8	$\frac{15}{16}$	$\frac{1}{16} = 6\frac{1}{4}\%$, and this is 15 of them.
9	264	$5\% = \frac{1}{20}$, so multiply 13.2 by 20.
*10	15204 to 16804 (exact: 16004)	Round each number to two figures: 1900 + 6400 + 7800 = 16100.
11	280	Evenly spaced, so 5 × the middle term: 5 × 56.
12	259	GCD = 7 and LCM = 63 × 28 ÷ 7 = 252.
13	5	Alternate digits from the right: 30 − 25 = 5.
14	21.60	A pound is 16 ounces: double 1.35 four times.
15	77	(53 − 46)(53 + 46) = 7 × 99; divide by 9.
16	16	Dividing by $\frac{3}{4}$ is multiplying by $\frac{4}{3}$: 12 × $\frac{4}{3}$ = 16.
17	3.54	Scale by $\frac{3}{7}$: 8.26 × 6 ÷ 14.
18	55	Month by month: 30 + 25 = 55.
19	1785	Convert first: 2497 − 712.
*20	237440 to 262432 (exact: 249936)	$(500 - 8)(500 + 8) = 500^2 - 8^2$, so about $500^2 = 250000$.
21	2816	Tens digits add to 10, same ones digit: 6 × 4 + 4 = 28, then $4^2 = 16$.
22	−30	Each absolute value first: 1 − 1 − 3 − 27.
23	4091060	Three digits per group: fill the empty places with zeros.
24	−0.3125	$3.2 = \frac{16}{5}$; flip it and keep the sign.
25	$132\frac{6}{25}$	Same whole part, fractions sum to 1: 11 × 12 = 132, then $\frac{3}{5} \times \frac{2}{5}$.
26	11227	Near 100: 103 + 9 = 112, then 3 × 9 = 27: 11200 + 27.
27	9	n(n − 3) ÷ 2 = 6 × 3 ÷ 2.
28	$\frac{54}{59}$	Improper fractions: $\frac{27}{5} \div \frac{59}{10} = \frac{27}{5} \times \frac{10}{59} = \frac{54}{59}$.
29	27000	$25 = 5^2$, so factor out 5: 5 × (5405 − 5) = 5 × 5400.
*30	1195 to 1319 (exact: ≈ 1257.0)	Keep three figures of each: 1060000 ÷ 844 ≈ 1256.
31	64	Set x = y = 1: $(5 - 3)^6 = 2^6$.
32	2024	Same tens digit, ones add to 10: 4 × 5 = 20, then 4 × 6 = 24.
33	−0.45	$2\frac{2}{9} = \frac{20}{9}$; flip it: $\frac{9}{20} = 0.45$, still negative.
34	94	The differences are 2, 3, 4, ..., so the step into the 14th term is 14: 108 − 14.
35	$\frac{4}{15}$	$(26 - 2)/90 = \frac{24}{90}$, reduced.
36	3	Use remainders on division by 6: 2 + 3 × 4 − 5 = 9, which leaves 3.

#	Answer	Shortcut
37	7	Add the equations: the x terms cancel, leaving 6y = 42.
38	2220	Go through base 2: three bits per base-8 digit gives 10101000, then pair the bits from the right.
39	1914	Halve 44 and double 43.5: 22 × 87, a ×11 product.
*40	748 to 826 (exact: ≈ 786.8)	$78^3 = 474552$ and $79^3 = 493039$ bracket the leading digits 487133, so the root is about 790.
41	11111000	$8 = 2^3$, so each base-8 digit becomes 3 base-2 digits.
42	480	$P = \frac{5}{6} \times \frac{1}{4} \times R = \frac{5}{24} \times R$, so $R = \frac{24}{5} \times P$.
43	80	${}_5C_1 \times 2^4 \times 1 = 5 \times 16 \times 1$.
44	7208	Tens digits add to 10: $100 \times (8^2 + 2^2 + 2 \times 2) + 2 \times 2^2 = 7200 + 8$.
45	0	Eliminate x: 11y = 0.
46	105	$a = 6\times7 - 7\times(-9) = 105$ and $b = 6\times(-9) + 7\times7 = -5$.
47	8	Count by leading digit (1: 0, 2: 0, 5: 2, 7: 6).
48	$\frac{5}{3}$	With everything on one side the constant is 10; the product is c ÷ a = 10 ÷ 6.
49	2440	Multiply right to left, carrying 6s. In base 10: 150 × 4 = 600, which is 2440_6.
*50	−923630 to −835666 (exact: −879648)	112 − 248 = −136, so about 6500 × (−140) = −910000.
51	82	The differences are the primes 2, 3, 5, 7, ...; the step into the 10th term is the 9th prime, 23.
52	−2	It is the log of the quotient: $8 \div 128 = \frac{1}{16} = 4^{-2}$.
53	127	$\log_5 25 = 2$, so f(25) = 125 + 2.
54	671	Ten terms of a Fibonacci-type sequence add up to 11 × the 7th term: 11 × 61.
55	1	11_7 is one more than the base, so alternate signs on the digits from the right: 2 − 1 + 5 − 5 = 1, which leaves 1 on division by 8.
56	6833	$\frac{41}{60} = 0.68333...$: the digits settle into a repeating one.
57	$\frac{1}{2}$	${}_5P_2 = 2! \times {}_5C_2$, and ${}_5C_3 = {}_5C_2$.
58	5858	$(10a + b)^2 + (10b - a)^2 = 101(a^2 + b^2)$ with a = 7, b = 3: 101 × 58.
59	4	Dividing by 8 in base 9 works like casting out nines: the digits add to 20, and 20 ÷ 8 leaves 4.
*60	119 to 131 (exact: 125)	Everything as a power of 5: 6 − 12 + 9 = 3, and $5^3 = 125$.
61	36	Divide each term by 5!: 5 × 6 + 6.
62	0.25	$\sec^2 = 1 + \tan^2 = 1 + \frac{1}{3}$, so $\cos^2$ is its reciprocal and $\sin^2 = 1 - \cos^2$.
63	$19\frac{4}{9}$	7 of the 36 rolls: $\frac{7}{36} = 19\frac{4}{9}\%$.
64	9	$k^3 = 756 - 27 = 729$.
65	0	Inside first: g(4) = −2, then f(−2).
66	$32\frac{8}{9}$	111 = 3 × 37, so the 3 cancels into 27: $74 \times \frac{4}{9} = \frac{296}{9}$.
67	1	$9^5 \equiv 1 \pmod{22}$, so only the exponent's remainder 35 mod 5 = 0 matters.
68	3	The 3 cancels: $g(\frac{1}{3}) = 0$, then f(0).
69	−16	Inside first: $g(-\frac{5}{6}) = -6$, then 3 × (−6) + 2.
*70	7868 to 8695 (exact: ≈ 8281.8)	Take the square root twice: $\sqrt{7970981907} \approx 89300$, then $\sqrt{89300} \approx 299$; then add 7983.
71	1.5	The inverse's slope is the reciprocal, $a = \frac{3}{2}$, and $b = -a \times \frac{3}{8} = -\frac{9}{16}$.
72	3.8	Changing the output by 6 moves the input by $\frac{6}{5}$: 5 − 1.2.

#	Answer	Shortcut
73	340	$f''(x) = 36x^2 + 16$; substitute $x = 3$.
74	10	The components are 0 and 10.
75	14.75	Set $f(x) = 14$: $4x + 11 = 70$, so $x = \frac{59}{4}$.
76	352	Cubes: $512 - 216 + 64 - 8$.
77	4	$\sin^2$ averages $\frac{1}{2}$ over a whole number of half-periods: half of $4 \times 2\pi$.
78	64481201	$(400 + 1)^3 = 400^3 + 3 \times 400^2 \times 1 + 3 \times 400 \times 1^2 + 1^3 = 64000000 + 480000 + 1200 + 1$.
79	84663	$4 \times 2 = 8$, cross terms $4 \times 7 + 2 \times 9 = 46$, last $9 \times 7 = 63$: $80000 + 4600 + 63$.
*80	10564 to 11675 (exact: ≈ 11120.0)	$\pi r^2 h = 216\pi$ cu. in., and 1 cu. in. $= 2.54^3 \approx 16.4$ cu. cm: $216 \times 3.14 \times 16.4 \approx 216 \times 51.5$.

Practice Test 5 — the shortcut for every answer

#	Answer	Shortcut
1	−1244	The other way round, 2011 − 767 = 1244; the answer is its negative.
2	23	Short division, left to right; check with 9 × 23 = 207.
3	4980	Add in pieces: 1222 + 3700 = 4922, then + 58.
4	9.01	$4\frac{1}{10} = 4.1$; line up the decimal points.
5	$\frac{13}{4}$	$3.25 = 3\frac{1}{4}$; then 3 × 4 + 1 over 4.
6	960	Factor out 30: 30 × (47 − 15) = 30 × 32.
7	361	$19^2 = 20 \times 18 + 1^2 = 360 + 1$.
8	17	× and ÷ left to right before + and −: 30 + 27 − 40 = 17.
9	2025	Ends in 5: 4 × 5 = 20, then attach 25.
*10	119962 to 132588 (exact: 126275)	Round each number to two figures: 63000 + 61000 + 2800 = 126800.
11	54	111 = 3 × 37, so 333 ÷ 37 = 9, and 9 × 6 = 54.
12	45	86 = 2 × 43.
13	1938	Split 34: 57 × 30 + 57 × 4 = 1710 + 228.
14	24570	×15: 1638 plus half of it is 2457, then ×10.
15	264	16 terms with average (9 + 24)/2: 16 × 33 ÷ 2.
16	8	Largest minus smallest: 8 − 0.
17	$\frac{8}{21}$	$2.625 = \frac{21}{8}$; flip it.
18	−4	(15 − 17)(15 + 17) = −2 × 32; divide by 16.
19	84	Month by month: 30 + 30 + 23 = 83, plus 1 because both end days count.
*20	24743 to 27347 (exact: 26045)	120 × 220 = 26400, then add about 630.
21	$-\frac{2}{21}$	A number plus its additive inverse is 0: change the sign.
22	14.8	The middle term x + 2 is the average of the five: 74 ÷ 5.
23	48	Double both: 432 ÷ 9.
24	7	With an odd exponent $9^5 + 5^5$ is divisible by 9 + 5 = 14, so only 7 matters: it leaves 7.
25	$56\frac{2}{9}$	Same whole part, fractions sum to 1: 7 × 8 = 56, then $\frac{2}{3} \times \frac{1}{3}$.
26	112.5	Cross-multiply: 15 × 30 ÷ 4.
27	2.72	8.5 × 32 = 272 cents; move the decimal point two places.
28	2.8	$1/x = \frac{1}{2} - \frac{1}{7} = \frac{5}{14}$; turn it over.
29	400	$s^2\sqrt{3} \div 4 = 1600\sqrt{3} \div 4$.
*30	550 to 607 (exact: ≈ 578.3)	$\sqrt{400} = 20$ and $\sqrt{836} \approx 29$, so about 20 × 29 = 580.
31	723	Place values: 8×81 + 8×9 + 3.
32	4	Take the square root twice: $\sqrt{256} = 16$, then $\sqrt{16} = 4$.
33	76	The side is $\sqrt{361} = 19$; four of them.
34	16	The numerator is $2(49x^2 + 4) = 2(49x^2 - 4) + 4 \times 4$.
35	46	94 = 2 × 47, so the count is (2 − 1)(47 − 1).
36	45	225 ÷ 5 = 45.
37	$\frac{35}{3}$	Subtract the equations: the y terms cancel, leaving −x = −19.

#	Answer	Shortcut
38	484	$22 = 11 \times 2$, so it is 121×4.
39	$\frac{23}{500}$	$4\frac{3}{5} = \frac{23}{5}$; put 100 more in the denominator and reduce.
*40	399 to 440 (exact: ≈ 419.1)	That is 165 inches at 2.54 cm each: about $165 \times 2.5 = 412.5$.
41	91809	$(300 + 3)^2$: $3^2 = 9$, $2 \times 3 \times 3 = 18$, $3^2 = 9$: $90000 + 1800 + 9$.
42	629	$k^2 = 2^2 + 25^2 = 4 + 625$.
43	5	Solve the line for each variable: $h = 4 - y$ and $k = 4 - x$. The image is (12, 7).
44	25704	$1836 \times 10 + 1836 \times 4 = 18360 + 7344$.
45	$\frac{30}{11}$	xy stays 30: $30 \div 11$.
46	23	Single digits mean the same in every base, so compute as usual and write the result in base 6. In base 10: $3^3 - 3^2 - 3 = 15$, which is 23_6.
47	36	The coefficients add up to the value at $x = 1$: $(3 - 9)^2$.
48	3885	Multiply first, then subtract, all in base 9. In base 10: $530 \times 6 - 268 = 2912$, which is 3885_9.
49	20	${}_6P_3 = 120$ and ${}_6C_1 = 6$.
*50	103143 to 113999 (exact: ≈ 108571.3)	2857.14 is $\frac{20000}{7}$, so it is $20000 \times 38 \div 7 \approx 108571$.
51	4	y ÷ x stays 1.5.
52	−1	It is the log of the quotient: $12 \div 36 = \frac{1}{3} = 3^{-1}$.
53	11	$m = 9$ makes the first factor $\frac{74}{9}$, and 37 divides 74; the second is then $18 \div \frac{74}{9} = \frac{81}{37}$, so $n = 2$.
54	420	Cancel 4!: $8 \times 7 \times 6 \times 5 = 1680$, then divide by 4.
55	23	Fractional exponents: $\frac{3}{2} + \frac{8}{6} = \frac{17}{6}$.
56	36	Sides 6.5 and 2.5; difference of squares: $(6.5 - 2.5)(6.5 + 2.5) = 4 \times 9$.
57	10	Pairing the terms leaves $1 + 2 + ... + 4 = 10$ with the sign of the last term.
58	115.8	Over the denominator 5 the numerators 4, 15, 19, ... are Fibonacci-type, so the 8 of them add up to the 10th minus the 2nd: $594 - 15 = 579$; then divide by 5.
59	7	$65 \equiv 7 \pmod{29}$, and $7^{14} \equiv 1$ ($a^{(p-1)/2}$ is always ±1 for a prime p); one more factor of 7 gives 7.
*60	698369 to 771880 (exact: ≈ 735124.4)	$0.474747... = \frac{47}{99}$: $\frac{47}{99}$ of 1806 is about 857, and its square is about 735124.
61	0.25	$\sin \frac{11\pi}{6} = -\frac{1}{2}$ and $\cos \frac{2\pi}{3} = -\frac{1}{2}$.
62	8	$5\sqrt{3} = \sqrt{75}$, between $8^2 = 64$ and $9^2 = 81$.
63	23858	Same first two digits: $(151 + 8) \times 150 = 23850$, plus $1 \times 8 = 8$.
64	225	The reference angle is 45°; in quadrant III that is 180° + 45°.
65	32	Inside first: $g(2) = 4$, then $f(4)$.
66	$131\frac{5}{9}$	$111 = 3 \times 37$, so the 3 cancels into 27: $296 \times \frac{4}{9} = \frac{1184}{9}$.
67	20	First term ÷ (1 − ratio): $10 \div \frac{1}{2} = 10 \times 2$.
68	$-\frac{7}{2}$	Expand both sides: $-2x + 15 = -4x + 8$.
69	22	In base 10 the number is $1 \times 125 + 3 \times 5 + 4 = 144 = 12^2$, and 12 is 22 in base 5.
*70	2388 to 2638 (exact: ≈ 2513.3)	$\pi r^2 h = 5^2 \times 32 \times \pi = 800\pi \approx 800 \times 3.14$.
71	−54	Remainder theorem: substitute $x = -2$, working synthetically.
72	55224	$(235 - 1)(235 + 1) = 235^2 - 1$, and 235^2 is $23 \times 24 = 552$ followed by 25.
73	4	$f''(x) = -4 \cos 2x$, and $2x = \pi$ there.

#	Answer	Shortcut
74	-8	Vertex at $x = -b/(2a) = -4$, where $y = -4$.
75	9	Complete the square: the radius is half of 6, so the area is $3^2\pi$.
76	$\frac{4}{3}$	$\sin u/u \to 1$ as $u \to 0$, so the limit is $\frac{4}{3}$.
77	-45	$p = 5k$ and $q = 6k$, so $-1k = 9$ and $k = -9$.
78	0	30 is a multiple of 5, so 30^7 contains 5^7, and 5^2 divides that evenly.
79	16	Two sequences are interleaved, 1, 5, 12, 22, ... and 1, 4, 9, 16, ...: the hidden terms are 16.
*80	319200 to 352800 (exact: 336000)	$8.888... = \frac{80}{9}$ and $37.8 \div 9 = 4.2$: $80 \times 4.2 \times 1000 = 336000$.